EXPERIMENT – 1

🎯 Aim

To create tables (EMPLOYEE, DEPARTMENT), insert records, and perform basic SQL operations like insert, update, and delete in MariaDB.

📘 Theory

In MariaDB, data is stored in tables consisting of rows and columns.

Key Concepts:
1. CREATE TABLE

Used to create a new table.

```
CREATE TABLE table_name (...);
```

2. INSERT INTO

Used to insert records.

```
INSERT INTO table_name VALUES (...);
```

3. UPDATE

Used to modify data.

```
UPDATE table_name SET column=value;
```

4. DELETE

Used to delete records.

💻 SQL Queries

```
MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| 2cse15g2_0896      |
| companay           |
| information_schema |
| mysql              |
| performance_schema |
| phpmyadmin         |
| test               |
+--------------------+
7 rows in set (0.058 sec)
```

```
MariaDB [(none)]> USE 2CSE15G2_0896;
Database changed
```

1. Create Employee Table

```
MariaDB [2CSE15G2_0896]> CREATE TABLE EMPLOYEE(EMPNO INT(4) PRIMARY KEY,ENAME
VARCHAR(20) NOT NULL, JOB VARCHAR(20), MGR INT(4), HIREDATE DATE,SAL INT(10),COMM
INT(7),DEPTNO INT(2),FOREIGN KEY(DEPTNO) REFERENCES DEPARTMENT(DEPTNO));
Query OK, 0 rows affected (0.020 sec)

MariaDB [2CSE15G2_0896]> DESC EMPLOYEE;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| EMPNO    | int(4)      | NO   | PRI | NULL    |       |
| ENAME    | varchar(20) | NO   |     | NULL    |       |
| JOB      | varchar(20) | YES  |     | NULL    |       |
| MGR      | int(4)      | YES  |     | NULL    |       |
| HIREDATE | date        | YES  |     | NULL    |       |
| SAL      | int(10)     | YES  |     | NULL    |       |
| COMM     | int(7)      | YES  |     | NULL    |       |
| DEPTNO   | int(2)      | YES  | MUL | NULL    |       |
+----------+-------------+------+-----+---------+-------+
8 rows in set (0.020 sec)
```

2. Create Department Table

```
MariaDB [2CSE15G2_0896]> DESC DEPARTMENT;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| DEPTNO | int(2)      | NO   | PRI | NULL    |       |
| DNAME  | varchar(15) | NO   |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
2 rows in set (0.024 sec)
```

3. Insert Values into Department

```
MariaDB [2CSE15G2_0896]> INSERT INTO DEPARTMENT VALUES (10, 'RESEARCH');
Query OK, 1 row affected (0.002 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO DEPARTMENT VALUES (20, 'ACCOUNTING');
Query OK, 1 row affected (0.001 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO DEPARTMENT VALUES (30, 'SALES');
Query OK, 1 row affected (0.002 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO DEPARTMENT VALUES (40, 'OPERATIONS');
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [2CSE15G2_0896]> SELECT*FROM DEPARTMENT;
+--------+------------+
| DEPTNO | DNAME      |
+--------+------------+
|     10 | RESEARCH   |
|     20 | ACCOUNTING |
|     30 | SALES      |
|     40 | OPERATIONS |
+--------+------------+
4 rows in set (0.000 sec)
```

4. Insert Sample Employee Records

```
MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7499,'ALLEN','SALESMAN',7698,'1981-02-20',1600,300,30);
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7521,'WARD','SALESMAN',7698,'1981-02-22',1250,300,30);
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7566,'JONES','MANAGER',7839,'1981-04-02',2975,NULL,20);
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7654,'MARTIN','SALESMAN',7698,'1981-09-28',1250,1400,30);
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7698,'BLAKE','MANAGER',7839,'1981-05-01',2850,NULL,30);
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7782,'CLARK','MANAGER',7839,'1981-06-09',2450,NULL,20);
Query OK, 1 row affected (0.002 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7788,'SCOTT','ANALYST',7566,'1982-12-09',3000,NULL,40);
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7839,'KING','PRESIDENT',NULL,'1981-11-17',5000,NULL,20);
Query OK, 1 row affected (0.004 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7844,'TURNER','SALESMAN',7698,'1981-09-08',1500,0,30);
```

```
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7876,'ADAMS','CLERK',7788,'1983-01-12',1100,NULL,20);
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7900,'JAMES','CLERK',7698,'1981-12-03',950,NULL,30);
Query OK, 1 row affected (0.002 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7902,'FORD','ANALYST',7566,'1981-12-03',3000,NULL,20);
Query OK, 1 row affected (0.004 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7934,'MILLER','CLERK',7782,'1982-01-23',1300,NULL,10);
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> INSERT INTO EMPLOYEE VALUES
(7369,'SMITH','CLERK',7902,'1980-12-17',800,NULL,20);
Query OK, 1 row affected (0.002 sec)

MariaDB [2CSE15G2_0896]> SELECT*FROM EMPLOYEE;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  800 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2450 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1100 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000 | NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1300 | NULL |     10 |
+-------+--------+-----------+------+------------+------+------+--------+
14 rows in set (0.001 sec)
```

5. Create Employee_Master Table

```
MariaDB [2CSE15G2_0896]> CREATE TABLE EMPLOYEE_MASTER AS SELECT*FROM EMPLOYEE;
Query OK, 14 rows affected (0.011 sec)
Records: 14  Duplicates: 0  Warnings: 0
```

```
MariaDB [2CSE15G2_0896]> SELECT*FROM EMPLOYEE_MASTER;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  800 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2450 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1100 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000 | NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1300 | NULL |     10 |
+-------+--------+-----------+------+------------+------+------+--------+
14 rows in set (0.000 sec)
```

6. Delete Records where deptno = 10

```
MariaDB [2CSE15G2_0896]> DELETE FROM EMPLOYEE_MASTER WHERE DEPTNO = 10;
Query OK, 1 row affected (0.003 sec)

MariaDB [2CSE15G2_0896]> SELECT*FROM EMPLOYEE_MASTER;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  800 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2450 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1100 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000 | NULL |     20 |
+-------+--------+-----------+------+------------+------+------+--------+
13 rows in set (0.000 sec)
```

7. Update Salary by 10% for Dept 20

```
MariaDB [2CSE15G2_0896]> UPDATE EMPLOYEE_MASTER SET SAL=SAL*1.10 WHERE DEPTNO=20;
```

```
Query OK, 6 rows affected (0.003 sec)
Rows matched: 6  Changed: 6  Warnings: 0

MariaDB [2CSE15G2_0896]> SELECT*FROM EMPLOYEE_MASTER;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  880 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 3273 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2695 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5500 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1210 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3300 | NULL |     20 |
+-------+--------+-----------+------+------------+------+------+--------+
13 rows in set (0.000 sec)
```

8. Alter Table (increase salary size)

```
MariaDB [2CSE15G2_0896]> ALTER TABLE EMPLOYEE_MASTER MODIFY SAL DECIMAL(10,2);
Query OK, 13 rows affected (0.028 sec)
Records: 13  Duplicates: 0  Warnings: 0

MariaDB [2CSE15G2_0896]> SELECT*FROM EMPLOYEE_MASTER;
+-------+--------+-----------+------+------------+---------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+--------+-----------+------+------------+---------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  880.00 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600.00 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250.00 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 3273.00 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250.00 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2695.00 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5500.00 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500.00 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1210.00 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950.00 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3300.00 | NULL |     20 |
+-------+--------+-----------+------+------------+---------+------+--------+
13 rows in set (0.001 sec)
```

9. Drop Table employee_master

```
MariaDB [2CSE15G2_0896]> DROP TABLE EMPLOYEE_MASTER;
Query OK, 0 rows affected (0.004 sec)
```

📄 Conclusion

Thus, tables were created, data inserted, updated, and deleted successfully in MariaDB.